Akilan Sankaran

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Education

Harvard University 2025 – 2029 (Expected)

Math 55; Physics 16; Data Structures/Algs; ML for Materials @ Kozinsky Lab **(XC Functionals, Neural Network Potentials)**; ML/Econometrics Project Lead for Charles River Economics Labs and Machine Intelligence Community

Albuquerque Academy | *High School Student* 2018 – 2025

Weighted GPA: 4.68/4.00; Highest in Class. *Courses*: Probability Theory, Formal Logic, ODEs/PDEs, Comp. Arch.

Scientific Research and Development Experience

Computational Materials Research Intern, Sandia National Laboratories (U.S. DOE) 2024 – 2025

- Evaluated carrier capture recombination rates within GaN semiconductors using first-principles Density Functional Theory (DFT) methodologies, together with harmonic analysis and perturbation theory.
- Developed first Python-based simulation library to evaluate recombination rates under a transition-state theory.

Degradation of "Forever Chemicals" via Electronic Structure Calculations 2023 – 2025

- Harnessed semiclassical approximations, quantum-scale methodologies (DFT/MD), and hybrid methods to develop and implement electronic-level modeling of several Per- and Polyflouroalkyl Substances (PFAS).
- Demonstrated first mechanistic understanding of Vitamin B_{12}-catalyzed defluorination of several PFAS.

Research Science Institute (RSI) – Pilot-Wave Hydrodynamics Research (MIT) (Preprint) 2024 – 2025

- Studied macroscopic tunneling analogues for millimetric walking droplets in MIT's Applied Mathematics Lab.
- Developed operator formalisms to rationalize experimental trials of variable-topography droplet dynamics.

Transgenic *Drosophila* Line Generation via CRISPR knock-in Methodology 2024 – 2025

- Wet-lab (collab. Stanford); enhancer-trapped stable-stock generation via balancer-based crossing schema.

Alternative Framing Developments regarding the *abc*-conjecture 2021 – 2023

- Formalized analytically tractable quality metrics with functional similarities to that of the *abc*-conjecture; enumerated high-quality triples with greater scaling efficiency under algorithmic optimization frameworks.

Exploration and Analysis of Highly Divisible Numbers 2020 – 2021

- Created alternative descriptions of highly composite numbers using number-theoretic smoothness considerations.

Technical Development Projects

"First Proof": Comparison of LLM Capabilities for Research-Level Mathematics (Preprint) Spring 2026

- Contrasted proof strategies of frontier models in the "First Proof" challenge, focusing on proof strategy.

***Slideo*: An Integrated Approach for General Speech-to-Command Slide Generation** Fall 2025

- As lead, designed and developed an end-to-end, fully automated, AI-based plugin for presentation generation and customization with Google Slides. Integrated with Google Chrome and OpenAI Whisper API for general use.

Alternatives to Standard Topic Modeling Approaches 2022 – 2023

- Investigated tailored alternatives to traditional topic-modeling algorithms within natural language processing.

International and National-level Scientific Recognition

- **Regeneron ISEF**: Two-time Top Mathematics and Chemistry Grand Awards (2022, 2024).
- **Coolidge Scholar**: Full-ride, presidential scholarship (5 out of over 4000 students).
- **Regeneron Science Talent Search (STS)**: Top 40 Finalist; **Broadcom MASTERS**: Top National Award
- **Steven H. Strogatz Prize for Mathematics Communication**: Top Winner, Social Media (2023).
- **Research Science Institute (RSI)**: Top 5 oral presentation out of ≈ 100 students, selected from > 3000.
- **USA Math Talent Search**: Top 7 scorers nationally; two-time gold medalist; **AIME**: 4-time qualifier.

Technical Skills

Languages/Tools: Python, LaTeX, MATLAB, Mathematica, Torch, JS, C++, Bash, Blender, VESTA, Avogadro, CFD Tools
ML/Math: ODE/PDE's, Fourier/functional analysis, optimization, measure/Lebesgue/representation theory; NLP, Kalman filters, numerical solvers, Speech-to-Text Models, Topic Models, LSTM, Bayesian processes, stochastic modeling, formal verification
Computational Chemistry/Biology: *ab-initio* modeling, quantum chemistry codes (Quantum Espresso/LAMMPS/VASP), ASE/Symbio, ML Force-Fields and Functionals, mechanistic reaction analysis, electron–phonon coupling, perturbation theory.